Practical No 4

Title of the Assignment: Implement BFS for solving the 8-puzzle problem

Problem statement: Use BFS to solve a planning problem. Examples can be 8 puzzle problem, Robot path planning, Blocks world etc.

Objective:

➢ To understand the concept of Uninformed (Blind) search techniques.

➢ To understand Breadth First Search (BFS) for solving a planning problem.

➢ To implement BFS for solving the 8-puzzle problem and obtain the solution path.

Theory:

Breadth First Search (BFS):

Breadth First Search (BFS) is an uninformed search technique that explores a state space level by level. It starts from the initial state and first explores all states at depth 1, then all states at depth 2, and so on until the goal state is reached. BFS uses a queue (FIFO – First In, First Out) to store the states that are waiting to be explored. A visited set is maintained to avoid revisiting the same state.

For a planning problem, each state represents a possible configuration of the problem, and each valid action generates a new state. BFS systematically explores these states until it finds a sequence of actions that transforms the initial state into the goal state.

Key characteristics of BFS:

➢ BFS is an uninformed (blind) search technique.

➢ It explores nodes level by level.

➢ It uses a FIFO queue.

➢ It does not require a heuristic function.

➢ BFS is complete when the branching factor is finite.

➢ For unit-cost actions, BFS finds a solution with the minimum number of steps.

BFS for 8-Puzzle:

In the 8-puzzle problem, a state consists of eight numbered tiles and one blank space arranged in a 3×3 grid. The blank tile can move up, down, left, or right whenever the move is valid. Each valid movement produces a new state. BFS treats every valid board configuration as a node in the search space and every blank-tile movement as an edge.

Initial state → BFS explores all possible states level by level → Goal state

The solution path is obtained by storing the parent of every newly generated state. When the goal state is reached, the parent links are followed backward to reconstruct the sequence of moves from the initial state to the goal state.

Algorithm:

Step 1: Start.

Step 2: Define the initial state and the goal state.

Step 3: Create an empty FIFO queue and a visited set.

Step 4: Insert the initial state into the queue and mark it as visited.

Step 5: Remove the front state from the queue.

Step 6: If the selected state is the goal state, reconstruct and display the solution path, then stop.

Step 7: Otherwise, generate all valid successor states by moving the blank tile.

Step 8: For each successor state, check whether it has already been visited.

Step 9: If the state is not visited, mark it as visited, store its parent, and insert it at the rear of the queue.

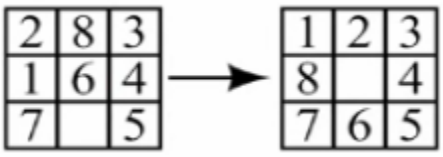
Step 10: Repeat Steps 5–9 until the goal state is reached or the queue becomes empty.

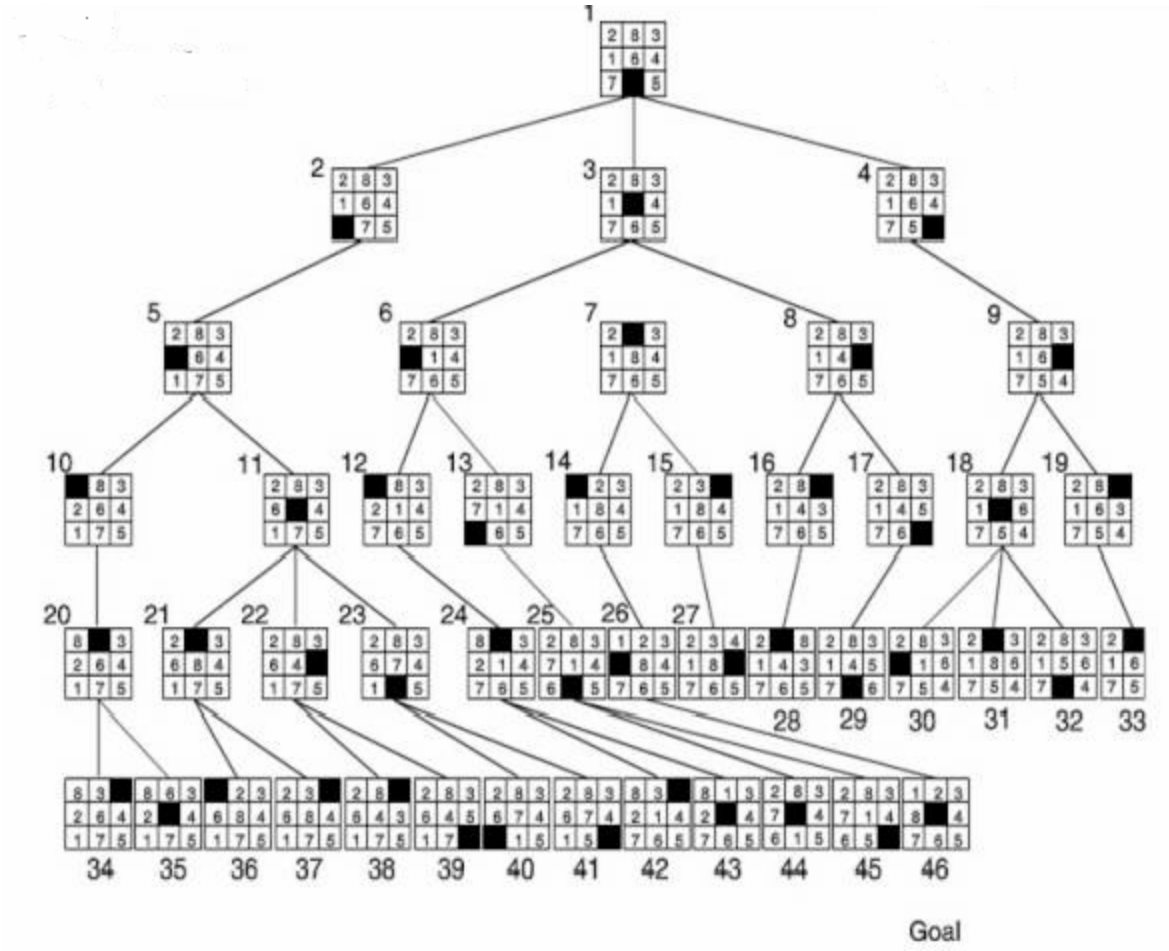
Step 11: If the queue becomes empty without finding the goal state, report that no solution was found.

Step 12: Display the sequence of states/moves from the initial state to the goal state.

Step 13: Stop.

8-Puzzle Problem:





Advantages and Limitations of BFS:

Advantages:

➢ Simple and systematic search strategy.

➢ Does not require heuristic information.

➢ Complete for a finite branching factor.

➢ Finds the shortest solution in terms of number of steps when all actions have equal cost.

Limitations:

➢ Requires large memory because many generated states may need to be stored.

➢ Can become computationally expensive for large search spaces.

➢ It does not use domain-specific knowledge to guide the search.

Conclusion:

BFS was studied and applied to a planning problem using the 8-puzzle example. The algorithm explores the state space level by level using a FIFO queue and avoids repeated states using a visited set. By storing parent information, BFS can reconstruct the sequence of actions required to reach the goal state..

Oral Questions:

1. What is Breadth First Search (BFS)?
2. Is BFS an informed or uninformed search technique?
3. What data structure is used by BFS?
4. What is the difference between BFS and DFS?
5. How is BFS applied to the 8-puzzle problem?
6. Why is a visited set required in BFS?
7. Is BFS complete? Explain.
8. When does BFS find the shortest path?
9. What are the applications of BFS in AI and planning?
10. What are the main limitations of BFS?